



Preface

Windows desktop applications continue to play a critical role in many scenarios among consumers and enterprises. Many of them have been the outcome of very long development processes that involved technologies that are now outdated, hard to maintain, and hard to integrate with the latest Windows innovations.

In this book, we'll show how you can modernize the UI and the features of your application using the Windows App SDK and WinUI, the latest iteration of the Windows developer platform.

Who this book is for

This book is for developers who are building Windows applications with Windows Forms, WPF, and UWP with a special focus on **Line-of-Business (LoB)** apps for enterprises. Basic knowledge of Windows app development, .NET/C#, and Visual Studio will assist with understanding the concepts covered in this book.

What this book covers

Chapter 1, *Getting Started with the Windows App SDK and WinUI*, covers the current status of the developer platform ecosystem and explains how you can start a new Windows App SDK project and how you can integrate its features into an existing Windows application.

Chapter 2, The Windows App SDK for a Windows Forms Developer, covers Windows Forms, which is one of the most popular development platforms for Windows. In this chapter, you'll learn how many concepts from Windows Forms are translated to the Windows App SDK and WinUI.

Chapter 3, The Windows App SDK for a WPF Developer, covers WPF, which has many similarities with the Windows App SDK and WinUI, starting from the shared UI layer based on XAML. In this chapter, you'll learn the key differences between WPF and WinUI.

Chapter 4, The Windows App SDK for a UWP Developer, explains that the Windows App SDK and WinUI are the direct successors to the UWP ecosystem. However, they are based on a different development model. In this chapter, we'll learn the differences we have to take into account.

Chapter 5, Designing Your Application, explains that building the UX of your application in the right way is critical to achieving success. This chapter will focus on important design elements, such as navigation, animations, and responsive layouts.

Chapter 6, Building a Future-Proof Architecture, explains that Windows desktop applications are often built to support critical tasks for many years to come. As such, building a future-proof architecture is critical to ensure you don't have to rebuild your application from scratch every time there's a requirement change. In this chapter, we'll learn how the MVVM pattern can help you build applications that are easy to maintain, test, and evolve over time.

Chapter 7, Migrating Your Windows Applications to the Windows App SDK and WinUI, puts into practice all the knowledge we acquired in the previous chapters to migrate existing Windows applications built with Windows Forms, WPF, and UWP to the Windows App SDK and WinUI.

Chapter 8, Integrating Your Application with the Windows Ecosystem, explains that Windows includes many innovative features that can be integrated into your application, such as Windows Hello and Geolocation. In this chapter, we'll learn how to use them in your Windows applications.

Chapter 9, Implementing Notifications, covers notifications, which play a key role in modern applications, delivering critical information to users. As such, Windows includes a powerful notification ecosystem, which you can integrate into your applications.

Chapter 10, Infusing Your Apps with Machine Learning Using WinML, explains that machine learning and artificial intelligence are two of the most important topics in the technology ecosystem. Windows enables developers to turn their machines into part of the edge ecosystem by supporting the evaluation of machine learning models without an internet connection and using the full power of the device. In this chapter, we'll learn how to integrate this platform, called WinML, into your Windows applications.

Chapter 11, Publishing Your Application, explains that developing a great application isn't enough to be successful. You have also to make it available to your users so that they can acquire it in the easiest possible way. In this chapter, we're going to explore the deployment opportunities offered by MSIX and platforms such as the Microsoft Store and App Installer.

Chapter 12, Enabling CI/CD for Your Windows Applications, covers DevOps, which is one of the key approaches nowadays to be successful in software development. In this chapter, we'll learn how you can use two of the key DevOps pillars, continuous integration and continuous deployment, to deliver your Windows applications in a more reliable, fast, and effective way.

To get the most out of this book

To build Windows desktop applications with the Windows App SDK and WinUI, you will need the following:

Software/hardware covered in the book	Operating system requirements
WPF	Windows
Windows Forms	Windows
Windows App SDK 1.0	Windows
WinUI 3.0	Windows
.NET 6	Windows
Visual Studio 2022	Windows

To follow the last chapter of the book, you will need also a GitHub account.

If you are using the digital version of this book, we advise you to type the code yourself or access the code from the book's GitHub repository (a link is available in the next section). Doing so will help you avoid any potential errors related to the copying and pasting of code.

Download the example code files

You can download the example code files for this book from GitHub at <https://github.com/PacktPublishing/Modernizing-Your-Windows-Applications-with-the-Windows-Apps-SDK-and-WinUI>. If there's an update to the code, it will be updated in the GitHub repository.

We also have other code bundles from our rich catalog of books and videos available at <https://github.com/PacktPublishing/>. Check them out!

Download the color images

We also provide a PDF file that has color images of the screenshots and diagrams used in this book. You can download it here:

https://static.packt-cdn.com/downloads/9781803235660_ColorImages.pdf.

Conventions used

There are a number of text conventions used throughout this book.

Code in text: Indicates code words in text, database table names, folder names, filenames, file extensions, pathnames, dummy URLs, user input, and Twitter handles. Here is an example: "You must use this initialization code before setting the **Source** property in XAML or calling the **EnsureCoreWebView2Async()** method."

A block of code is set as follows:

```
<VariableSizedWrapGrid Orientation="Horizontal"
MaximumRowsOrColumns="3" ItemHeight="200"
ItemWidth="200">
    <Rectangle Fill="Red" />
    <Rectangle Fill="Blue" />
    <Rectangle Fill="Green" />
    <Rectangle Fill="Yellow" />
</VariableSizedWrapGrid>
```

Any command-line input or output is written as follows:

```
winget install Microsoft.Edge
```

Bold: Indicates a new term, an important word, or words that you see onscreen. For instance, words in menus or dialog boxes appear in **bold**. Here is an example: "Click on **Upload files** and wait for the process to finish."

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